

DEV JUNEJA

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AI/ML Engineer in training, building RAG, fine-tuning, and transformer-based projects; full-stack developer shipping production-ready apps.

EDUCATION

Vellore Institute of Technology (VIT Bhopal)

Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering (AI & ML) | CGPA: 7.75

Bhopal, MP

PROJECTS

RLM-Python Recursive Language Models | Python, Gemini API, REPL Sandbox | [GitHub](#)

- Built an independent Python implementation of **Recursive Language Models (RLM)**, treating long context as external memory instead of stuffing it into every prompt
- Designed a **Python REPL sandbox** where a root LLM programmatically inspects, slices, and searches a stored context, dispatching recursive sub-LLM calls (`llm_query`, `llm_query_parallel`) only on relevant snippets
- Implemented an async engine with Gemini-based root/sub drivers and a **Needle-in-a-Haystack benchmark** to evaluate long-context retrieval accuracy

Brainly AI Second Brain | MERN, TypeScript, Tailwind, JWT, pgvector, OpenAI | [Live Demo](#) | [GitHub](#)

- Built a full-stack **AI-powered knowledge base** to save YouTube videos, tweets, and links with semantic search via **RAG (Retrieval-Augmented Generation)**
- Integrated **pgvector** with PostgreSQL to store and query embeddings; content is chunked, embedded, and retrieved contextually to answer natural language queries over saved notes
- Architected a **monorepo** with React + Tailwind frontend and Node.js + Express backend; implemented **JWT auth**, RESTful APIs, and Mongoose schemas

MiniLLM | Python, PyTorch, BPE Tokenizer | [Colab](#)

- Built a miniature transformer LLM from scratch in PyTorch, trained on Shakespeare text on a Google Colab T4 GPU
- Implemented modern architectural components including **RoPE positional encoding**, **Grouped Query Attention (GQA)**, **SwiGLU** activations, and **RMSNorm** mirroring techniques used in LLaMA-style models
- Engineered a custom **Byte Pair Encoding (BPE) tokenizer**; diagnosed and resolved garbled output traced to corpus size limitations, improving generation quality significantly

SKILLS

Languages : Python, JavaScript, TypeScript, C++, SQL

Frameworks & Libraries : React.js, Node.js, Express.js, Next.js, FastAPI, PyTorch, Scikit-learn, HuggingFace Transformers, LangChain

Databases & ORMs : MongoDB, PostgreSQL, Prisma, Mongoose, pgvector

Tools & Platforms : Git, GitHub, Docker (basics), Postman, Google Colab, VS Code, GitHub Actions

AI & ML : Transformer Architecture, RAG, Fine-tuning, BPE Tokenization, Embeddings, Agents, RoPE, GQA, SwiGLU

Concepts : REST APIs, WebSockets, JWT Auth, DSA (Striver A2Z)

LEADERSHIP & INVOLVEMENT

Data Science Club, VIT Bhopal

Aug. 2025 – July 2026

Technical Team Lead

Bhopal, MP

- Leading the technical wing of the club; mentoring junior members on ML/DL concepts and project development
- Organising workshops, hands-on sessions, and club projects to build a data science learning community

Data Science Club, VIT Bhopal

Aug. 2024 – Jul. 2025

Technical Team Member

Bhopal, MP

- Contributed to club technical initiatives, assisted in event planning, and collaborated on internal ML projects